

Final Project

The course project will involve engagement with a current topic of research in algorithmic game theory or mechanism design. You can choose to do the project by yourself or with a partner. You should strive to produce something novel, such as a new algorithm, model, theorem, or counterexample. All projects this semester are theoretical research projects. Reading related work is part of the research process; a survey alone does not fulfill the project requirement.

The project involves several deliverables:

1. (8 points) A proposal describing related work and the intended topic and scope of the project.
2. (27 points) A brief in-class presentation.
3. (45 points) A final technical report, typeset in LaTeX, describing the problem you considered, related work, your approach, your results, and open questions.

Your project proposal should be **submitted on Gradescope by 11:59pm on September 30**, and should include:

- (2 points) Names of group members (1–2 people).
- (2 points) A description of your research question, the model you plan to study, and the most relevant related work.
- (2 points) A brief work plan, including dates for intermediate tasks and a first special case or lemma to work on.
- (2 points) For pairs, a description of how you will split the work and coordinate (e.g., regular meeting times). For individual projects, a description of how you will organize your work and check your progress.

Presentations will be in class on **December 3, 8, and 10**. Your final write-up should be **at least 6 pages, not including references**, and submitted on Gradescope by the final-exam date and time listed on the course website. Include a summary of existing work, your new results with proofs, a discussion of challenges faced and other approaches attempted, and suggestions for future work.

Original Research

Your project should solve, or begin to solve, a theoretical research question beyond the material covered in the lectures. A useful first step is to take a model or mechanism from a paper and change one assumption: what the agents know, what they can report, what the designer observes, or what outcomes are feasible. Start with a small instance that you can analyze completely. A proved special case, an impossibility example, or a rigorous counterexample to a natural approach can be meaningful progress. Reproducing a known result can help you get started, but the project should go beyond reproduction.

If you are interested in working together with me on a research project, I strongly encourage you to talk to me one-on-one. We can discuss ongoing or early-stage projects that are part of my own agenda. I am actively looking for students and collaborators, and I have many research problems that I'm happy to share.

Suggested Topics and Starting Points

The topics below extend beyond the lecture material. Each calls for a new model with mathematical analysis, an algorithm with proved guarantees, a characterization, or an impossibility result. These are starting points for developing a project, not claims that every variant is open. Read the related work and discuss a precise question with me before committing to it. You are welcome to propose another topic related to the course. You do not need to wait until we cover a topic in class to begin reading about it.

Mechanism Design for Generated Content. Study a mechanism whose outcome is a generated sequence rather than an allocation of a fixed item. How do token-by-token incentive guarantees relate to incentives over the complete output when values depend on combinations of tokens? Choose a restricted valuation class and prove conditions under which local payments induce truthful behavior, or give an impossibility result. This introduces a sequential outcome space beyond the delegated bidding model from class.

- Paul Dütting, Vahab Mirrokni, Renato Paes Leme, Haifeng Xu, and Song Zuo. “Mechanism Design for Large Language Models.” WWW 2024.

Mechanism Design with Predictions. Study strategyproof facility location with an external prediction. Choose a preference domain or prediction-error model beyond the guarantees in the papers below, and prove a tradeoff between performance with accurate predictions and worst-case performance. This adds a prediction-dependent mechanism and an error-dependent benchmark to the prior-independent auction models from class.

- Priyank Agrawal, Eric Balkanski, Vasilis Gkatzelis, Tingting Ou, and Xizhi Tan. “Learning-Augmented Mechanism Design: Leveraging Predictions for Facility Location.” Mathematics of Operations Research 2024.
- Eric Balkanski and Cherlin Zhu. “Strategyproof Learning with Advice.” ALT 2025.

Algorithmic Contract Theory. A principal pays an agent based on an observable outcome, but cannot observe the costly action that produced it. Study the power of a restricted class of contracts when the agent has several actions or the principal has limited information. Prove an approximation guarantee relative to the optimal contract, a lower bound, or a characterization for a restricted action structure.

- Paul Dütting, Tim Roughgarden, and Inbal Talgam-Cohen. “Simple versus Optimal Contracts.” EC 2019.

Correlation-Robust Mechanism Design. Suppose a seller knows marginal value distributions but not their correlation. Evaluate a mechanism by its worst-case expected revenue over compatible joint distributions. How does additional information restricting the possible correlations change the optimal mechanism? Choose a small multidimensional environment and seek an optimality characterization or a bound on the value of that additional information.

- Gabriel Carroll. “Robustness and Separation in Multidimensional Screening.” *Econometrica* 2017.
- Moshe Babaioff et al. “Escaping Cannibalization? Correlation-Robust Pricing for a Unit-Demand Buyer.” EC 2020.

Selling Information. A buyer’s value for information depends on the decision it helps them make. Study a seller who can offer different signals about an unknown state at different prices. For a restricted decision problem, characterize an optimal menu of signals or bound the revenue loss from restricting the menu’s size. Identify a variant beyond the existing results, such as an additional restriction on disclosure or a different class of buyer types.

- Yang Cai and Grigoris Veleghkas. “How to Sell Information Optimally: An Algorithmic Study.” ITCS 2021.
- Dirk Bergemann, Yang Cai, Grigoris Veleghkas, and Mingfei Zhao. “Is Selling Complete Information (Approximately) Optimal?” EC 2022.

Tournament Rules for Larger Coalitions. The lecture studies manipulation by pairs. Study coalitions of three or more teams, which can change several internal match outcomes at once. For a specified family of tournament rules, prove an upper or lower bound on the coalition’s possible gain, or study how requiring a stronger winner-selection property changes that bound. Begin with the larger-coalition results below and identify a restricted case that remains unresolved.

- Jon Schneider, Ariel Schwartzman, and S. Matthew Weinberg. “Condorcet-Consistent and Approximately Strategyproof Tournament Rules.” ITCS 2017.
- Ariel Schwartzman, S. Matthew Weinberg, Eitan Zlatin, and Albert Zuo. “Approximately Strategyproof Tournament Rules: On Large Manipulating Sets and Cover-Consistence.” 2019 preprint.

Transaction Fee Mechanisms. In a blockchain, the party selecting transactions may itself act strategically, insert fictitious transactions, or coordinate with users. Choose a restricted model of block capacity and allowable deviations. Can a fee rule simultaneously satisfy user incentive compatibility, miner incentive compatibility, and resistance to side agreements? Prove a tradeoff or design a mechanism under a clearly specified restriction.

- Tim Roughgarden. “Transaction Fee Mechanism Design.” JACM 2024.
- Tim Roughgarden. “Transaction Fee Mechanism Design for the Ethereum Blockchain: An Economic Analysis of EIP-1559.” 2020 report.

Fair Allocation under Uncertain Demand. Allocate a fixed supply across groups before their demands are realized. How much utilization must be sacrificed to guarantee comparable service probabilities when the demand distributions are only partially known? Specify an ambiguity set of distributions and prove a fairness–utilization bound or a matching lower bound. This introduces distributional uncertainty into a resource-allocation problem beyond the health-insurance market-entry model from class.

- Kate Donahue and Jon Kleinberg. “Fairness and Utilization in Allocating Resources with Uncertain Demand.” FAT* 2020.

Mechanism Design with Unknown Present Bias. In a task-graph model, suppose the designer knows only an interval containing the agent’s present-bias parameter. Can one reward or commitment scheme motivate every type in that interval? Starting from the uncertainty model below, change the permitted interventions or the information available to the designer and seek a new bound on the cost of robustness. The project should address uncertainty about the agent’s behavioral parameter, beyond tracing a known type’s choices in class.

- Jon Kleinberg and Sigal Oren. “Time-Inconsistent Planning: A Computational Problem in Behavioral Economics.” EC 2014.
- Susanne Albers and Dennis Kraft. “The Price of Uncertainty in Present-Biased Planning. WINE 2017.